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Physics | Lecture - 12

Basic Maths and Calculus

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Todays Goal

PhD. on integration



Join it

(8) Velocity of object $V = \frac{t^3}{3} - 4t^2 + 12t$ then find Maxm & minimum velocity of object

Soln

$$V = \frac{t^3}{3} - 4t^2 + 12t$$

It will maxm or minm when $\frac{dV}{dt} = 0$

$$\frac{1}{3}(3t^2) - 4(2t) + 12 \times 1 = 0$$

$$t^2 - 8t + 12 = 0$$

$$t^2 - 6t - 2t + 12 = 0$$

$$t(t-6) - 2(t-6) = 0$$

$$(t-6)(t-2) = 0$$

$t_1 = 6$ $t_2 = 2$

$$V_{(t_1=6)} = \frac{216}{3} - 4(36) + 12 \times 6$$

min at $(t_1=0)$ $= 72 - 144 + 72$

$= 0$

$$V_{t=2} = \frac{8}{3} - 4(4) + 12 \times 2$$

$$= \frac{8}{3} - 16 + 24 = \frac{8}{3} + 8$$

$$= \frac{32}{3}$$

V_{max} (at $t=2$) $= +\frac{32}{3}$

(Q) $\frac{d(\sin^4\theta)}{d(\sin\theta)} = ??$

$\frac{d\sin^4\theta}{d\theta} = \frac{d(\sin\theta)^4}{d\theta} = 4(\sin\theta)^{4-1}\frac{d\sin\theta}{d\theta}$

$= 4(\sin\theta)^3 \times \cos\theta$

(a) $4\sin^3\theta \cdot \cos\theta$ (27·1·)

(b) $4\sin^3\theta$ (45·1·)

(c) $\cos^4\theta$ (13·1·)

(d) $-4\sin^3\theta\cos\theta$ (15·1·)

$\sin\theta = x$ (let)

$\frac{dx^4}{dx} = 4(x)^3 = 4(\sin\theta)^3$

$\frac{dx^4}{dx} = 4x^3$

(Q) $$\frac{d \log_e(\sin\theta)}{d(\sin\theta)}$$

Let $\sin(\theta) = x$

$$\frac{\log_e x}{dx} = \frac{1}{x} = \frac{1}{\sin\theta}$$

(a) $\frac{1}{\sin\theta}$

(b) $\cot\theta$

(c) $\tan\theta$

(d) $\frac{1}{\cos\theta}$

find

HW

$$\frac{d(x)^4}{d(x)^2} = ??$$

(a) $4x^3$

(b) $2x^2$

(c) $4x^{3/2}$

$x^2 = t$

$$\frac{dt^2}{dt} = 2t = 2x^2$$



$$F_{gravitational} = \frac{GMm}{r^2}$$

⊙ Newtons law of gravitation only valid for Point spherical mass



$$F_{gravitn} = \frac{GMm}{(r+\frac{l}{2})} \times$$

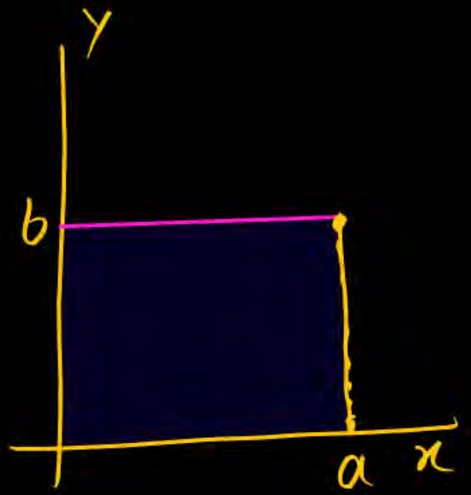
$$dF = \frac{GM\,dm}{x^2}$$

Integration → A mathematical operator use to add large no of infinitly small term.

Area under curve

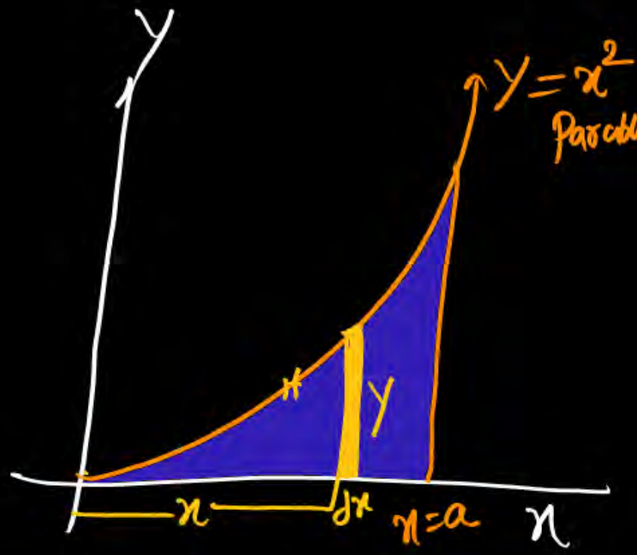


Area under curve $= \frac{1}{2}ab$



Area under curve on x-axis

Area $= ab$



$$\int dA = \int y\,dx$$

Area under curve $= \int y\,dx$ *

integration

MR* Box

Slope of y/x graph $= \frac{dy}{dx}$

= Rate of change in y w.r.t. x

Area of y/x graph $= \int y\,dx$

Integration → method to add large no. of infinitly small terms.
→ Inverse of differentiation.

$\int y\,dx$ = integration of y w.r.t x

$\int a\,dt$ = intⁿ of a w.r.t 't'

$\int P\,dt$ = integⁿ of P w.r.t 't'

$\int y^2\,dt$ = intⁿ of y^2 w.r.t 't'

$\int V\,dt$ = intⁿ of V w.r.t t

integration

- indefenite integ^n (No limit)

$$\int y\,dx = [\text{Result}] + C$$

definite integ^n (with limit)

$$\int_{x_1}^{x_2} y\,dx = (\text{Result})_{x_1}^{x_2}$$

$$= \text{Result}(x_2) - \text{Result}(x_1)$$

Rule of Integration

Rule-1

$y = x^n$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

n = power but $n \neq -1$

Ex-

$$\int x^2 \, dx = \frac{x^{2+1}}{2+1} + C = \frac{x^3}{3} + C$$

$$\Rightarrow \int x^3 \, dx = \frac{x^{3+1}}{3+1} = \frac{x^4}{4} + C$$

$$\int \sqrt{x} \, dx = \int x^{1/2} \, dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{x^{3/2}}{3/2}$$

$$\int x^{2/3} \, dx = \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1} + C = \frac{x^{5/3}}{5/3} = \frac{3}{5} x^{5/3}$$

$$\int x^{3/5} \, dx = \frac{x^{\frac{3}{5}+1}}{\frac{3}{5}+1} + C$$

$$\int \frac{1}{x^2} \, dx = \int x^{-2} \, dx = \frac{x^{-2+1}}{-2+1} = \frac{x^{-1}}{-1} = -\frac{1}{x}$$

$$\int \frac{1}{\sqrt{x}} \, dx = \int x^{-1/2} \, dx = \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{x^{1/2}}{1/2}$$

$$= 2\sqrt{x}$$

$$\# \quad \int \frac{1}{x}\,dx = \int x^{-1}\,dx = \frac{x^{-1+1}}{-1+1} = \frac{x^0}{0} + c$$ ✗

Rule-2

integration

$$\int \frac{1}{x}\,dx = \log e^x$$

diff^n

Rule-3

diff^n

$$\int \cos\theta\,d\theta = \sin\theta + c$$

int^g

$$\int \sin\theta\,d\theta = -\cos\theta + c$$

$$\int \sec^2\theta\,d\theta = \tan\theta + c$$

$$\int \tan\theta \cdot \sec\theta\,d\theta = \sec\theta + c$$

Rule-4

$$\int e^x \, dx = e^x + C$$

diff^n

Rule-5 $y = x^0 = 1$

$$\int x^0 \, dx = \frac{x^{0+1}}{0+1} = \frac{x^1}{1} = x$$

$$\int 1 \, dx = \int dx = x$$

- $\int dt = t$
- $\int dA = A$
- $\int dP = P$
- $\int d(\text{Popu}) = \text{Popu.} + C$
- $\int d(xy) = xy$

$$\int 1 \, d(mR) = m \cdot R.$$

$$\int d\theta = \theta$$

Rule No-6 Integration of constant function

$y = 5$

$$\int y\,dx = \int 5\,dx = 5\int dx = 5x + c$$

$5x + c$ → diff^n w.r.t x

diff^n

$$\frac{d(5x+c)}{dx} = 5\frac{dx}{dx} + \frac{dc}{dx}^{0}$$

$$= 5$$

$y = 10$

$$\int y\,dt = \int 10\,dt = 10t + c$$

$y = 2$

$$\int y\,dx = \int 2\,dx = 2\int dx = 2x + c$$

$y = 25x^2$

$$\int y\,dx = \int 25x^2\,dx$$

$$= 25\int x^2\,dx$$

$$= 25\left[\frac{x^3}{3}\right] + C$$

$$= \frac{25}{3}x^3$$

Rule-7 Addition Rule / subtractor

\# $y = \frac{1}{4}x^3 + \frac{x^2}{3} + \frac{x}{2}$

$$\int \boxed{y}\, dx = \int \underbrace{\left(\frac{x^3}{4} + \frac{x^2}{3} + \frac{x}{2}\right)}_{y} dx$$

$$= \int \frac{x^3}{4}\, dx + \int \frac{x^2}{3}\, dx + \int \frac{x}{2}\, dx$$

$$= \frac{1}{4}\int x^3\, dx + \frac{1}{3}\int x^2\, dx + \frac{1}{2}\int x\, dx$$

$$= \frac{1}{4}\,\frac{x^4}{4} + \frac{1}{3}\,\frac{x^3}{3} + \frac{1}{2}\,\frac{x^2}{2} + c = \frac{x^4}{16} + \frac{x^3}{9} + \frac{x^2}{4} + c$$

$\int \frac{1}{x} = \log_e x + c$

$\int \frac{1}{x^2} dx = \int x^{-2} dx = \frac{x^{-2+1}}{-2+1} = \frac{x^{-1}}{-1} = -x^{-1} = \frac{-1}{x^1}$

$\int \frac{1}{x^4} dx = \int x^{-4} dx = \frac{x^{-4+1}}{-4+1} = \frac{x^{-3}}{-3} = -\frac{1}{3x^3}$

$\int \frac{1}{x^{10}} dx = \int x^{-10} dx = \frac{x^{-10+1}}{-10+1} = \frac{x^{-9}}{-9} = -\frac{1}{9x^9} + c$

$\int \frac{1}{x^{3/2}} dx = \int x^{-3/2} dx = \frac{x^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} + c$

$$\int \sqrt{x}\, dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}$$

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-\frac{1}{2}} dx = \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}$$

$$= \frac{x^{1/2}}{1/2}$$

$$= 2\sqrt{x}$$

Rule :-7

$$\Rightarrow \int (\sin x + e^x - 4x^2)\,dx = \int \sin x\,dx + \int e^x\,dx - 4\int x^2\,dx$$

$$= -\cos x + e^x - 4\frac{x^3}{3}$$

$$\Rightarrow \int (4t^3 - 3t^2 + 4)\,dt = 4\left[\frac{t^{3+1}}{3+1}\right] - 3\frac{t^3}{3} + 4t$$

$$= t^4 - t^3 + 4t$$

$$\Rightarrow \int (4t - 2 + e^t)\,dt$$

$$= 4\frac{t^2}{2} - 2t + e^t$$

$$= \left[2t^2 - 2t + e^t\right] + c$$

$$\int \sin x\,dx = -\cos x$$

$$\int \cos x\,dx = \sin x$$

Out-side Inside Rule

$$\# \quad \frac{dy}{dx} = \left[\begin{array}{l}\text{diff}^{n} \text{ of outer function} \\ \text{Keep inside as it is}\end{array}\right] \times \begin{array}{l}\text{diff}^{n} \text{ of} \\ \text{Inner} \\ \text{function}\end{array}$$

Rule-9

Integration Outside Inside Rule

$$\int y\,dx = \frac{\left[\begin{array}{l}\text{Integ}^{n} \text{ of outer function} \\ \text{Keep inside as it is}\end{array}\right]}{\text{Coefficient of } [x]}$$

$$\# \quad \int \sin(2\theta)\,d\theta = \frac{-\cos(2\theta)}{2} + c$$

$$\# \quad \int \cos(3x-5)\,dx = \frac{\sin(3x-5)}{3} + c$$

$$\# \quad \int \sin(5-4x)\,dx = \frac{+\cos(5-4x)}{+4} \checkmark$$

$y = \sin(2x+3)$

$$\int \sin(2x+3)\,dx = -\frac{\cos(2x+3)}{2}$$

$y = e^{(2x-4)}$

$$\int e^{(2x-4)}\,dx = \frac{e^{(2x-4)}}{2}$$

$y = (3x+2)^4$

$$\int (3x+2)^4\,dx = \frac{(3x+2)^{4+1}}{(4+1)\times 3}$$

$$= \frac{(3x+2)^5}{15} + c$$

\# $$\int (Ax+B)^2\,dx = \frac{(Ax+B)^3}{3\times A} + c$$

$$\int e^{-2x}\,dx = \frac{e^{-2x}}{-2} + c$$

$\int \sin(4x^2)\,dx =$ Not in syllabus

$\int \sin(4x)\,dx = -\frac{\cos(4x)}{4} + c$

$\int (5-8x)^{10}\,dx = \frac{(5-8x)^{11}}{11 \times (-8)} + c$

$\int e^{(5-3x)}\,dx = \frac{e^{(5-3x)}}{-3} + c$

Rule no

$\int \frac{1}{x}\,dx = \log_e x$

$\int \frac{1}{2x}\,dx = \frac{1}{2}\int \frac{1}{x}\,dx = \frac{1}{2}\log x$

$\int \frac{1}{(2x+3)}\,dx$

$$\Rightarrow \int (2x+3)^8 \, dx = \frac{(2x+3)^9}{9 \times 2} + C$$

$$\Rightarrow \int \frac{1}{(2x+3)^8} dx = \int (2x+3)^{-8} dx = \frac{(2x+3)^{-8+1}}{(-8+1) \times 2} = \frac{(2x+3)^{-7}}{(-7) \times 2}$$

Rule-No

$$\int \frac{1}{(2x+3)} dx = \int (2x+3)^{-1} dx \quad \times$$

$$= \frac{\log_e (2x+3)}{2} + c \quad \checkmark$$

(#) $$\int \frac{1}{(3x-4)}\,dx = \frac{\log_e(3x-4)}{3} + c$$

(#) $$\int \frac{1}{(B-Ax)}\,dx = \frac{\log(B-Ax)}{-A}$$

(#) $$\int \frac{1}{(4-8x)}\,dx = \frac{\log_e(4-8x)}{-8} + c$$

(#) $$\int \frac{1}{(4-8x)^2}\,dx = \frac{(4-8x)^{-2+1}}{(-2+1)\times(-8)} = \frac{(4-8x)^{-1}}{+8} = \frac{1}{8(4-8x)^1} + c$$ ✓

Find

$$\int_0^1 \frac{(x^2-a^2)}{(x+a)}\,dx$$

Integration with limit

$y = x^2 + 1$ integrate from $x_1 = 1$ to $x_2 = 2$

$$\int y\,dx = \int_1^2 (x^2+1)\,dx = \left(\frac{x^3}{3} + x\right)_1^2$$

$$= \left[\frac{(2)^3}{3} + 2\right] - \left(\frac{1^3}{3} + 1\right)$$

$$= \frac{8}{3} + 2 - \frac{1}{3} - 1$$

$$= \frac{8}{3} - \frac{1}{3} + 1 = \frac{8 - 1 + 3}{3} = \frac{10}{3}$$

Find

$$\frac{d}{dx}\int_0^1 \frac{(x^2-a^2)}{(x+a)}\,dx$$

where a is cotη

$$\frac{d}{dx}\int_0^1 \frac{(x-a)\cancel{(x+a)}}{\cancel{(x+a)}}\,dx$$

$$\frac{d}{dx}\int_0^1 (x-a)\,dx = 0$$

$$\frac{d}{dx}\left[\int x\,dx - \int a\,dx\right]$$

$$\frac{d}{dx}\left[\frac{x^2}{2} - ax\right]_0^1$$

$$\frac{d}{dx}\left[\frac{1}{2} - a\right] - 0$$

$$= \frac{d}{dx}\left(\frac{1}{2} - a\right) = 0$$

↑ cotη

$\int (\sin 30^\circ)\, d\theta = \int \frac{1}{2}\, d\theta = \frac{1}{2}\theta$ ✓

$\int e^2\, dx = e^2$ ✗

$\Rightarrow \int e^2\, dx = e^2 \int dx = e^2 x$ ✓

$\int e^n\, dx = e^x$ ✓

$\int e^5\, dx = e^5 x$ ✓

(#)

$$\int_0^1 e^x \, dx = \left[e^x\right]_0^1$$

$$= e^1 - e^0$$

$$= e - 1$$

$$= 2.71 - 1$$

$$= 1.71$$

Ans

$$\int_0^1 e^{-2x} \, dx = \left[\frac{e^{-2x}}{-2}\right]_0^1 = -\frac{1}{2}\left[e^{-2} - e^0\right]$$

$$= -\frac{1}{2}\left(\frac{1}{e^2} - 1\right) = -\frac{1}{2}\left[\frac{1 - e^2}{e^2}\right]$$

(##)

$$\int_{-1}^{+1} e^{-x}\,dx = \frac{e^{-x}}{-1} = \left[-e^{-x}\right]_{-1}^{1}$$

$$= -1\left[e^{-1} - e^{1}\right]$$

$$= -1\left[\frac{1}{e} - e\right]$$

$$= e - \frac{1}{e}$$

$$= \left[\frac{e^2-1}{e}\right]$$

$\int_0^{\pi/2} \sin(x)\,dx = \left[-\cos x\right]_0^{\pi/2}$

$= \cos(0) - \cos(\pi/2)$

$= 1 - 0 = \underline{+1}$

$\int_0^{3\pi/2} \sin\theta\,d\theta = 1$

$\pm \int_0^{\pi} \cos\theta\,d\theta = \left[\sin\theta\right]_0^{\pi} = \sin\pi - \sin 0$

$= 0 - 0 = 0$

$\int_0^{2\pi} \cos\theta\,d\theta = 0$

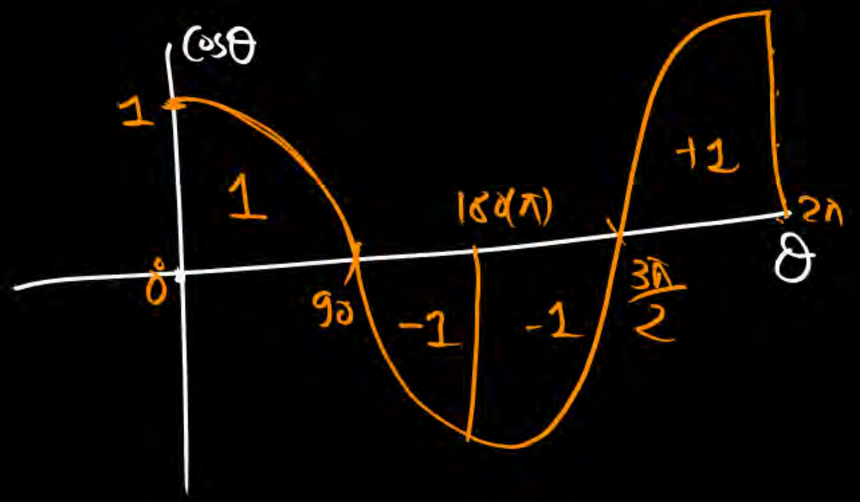
$\int_{\pi/2}^{\pi} \sin\theta\,d\theta = 1$

$$\int_{-\pi/2}^{+\pi/2} \cos\theta \, d\theta = 2$$

$$\int_{\pi}^{2\pi} \sin\theta \, d\theta = -2 \checkmark$$

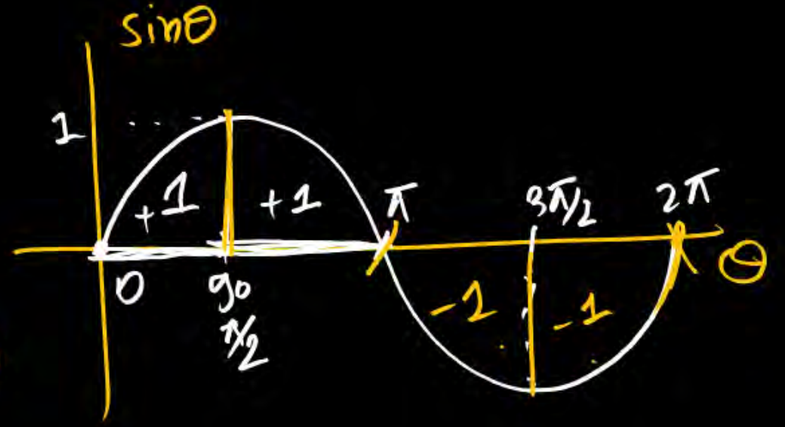
$$\int_{\pi/2}^{3\pi/2} \cos\theta \, d\theta = -2$$

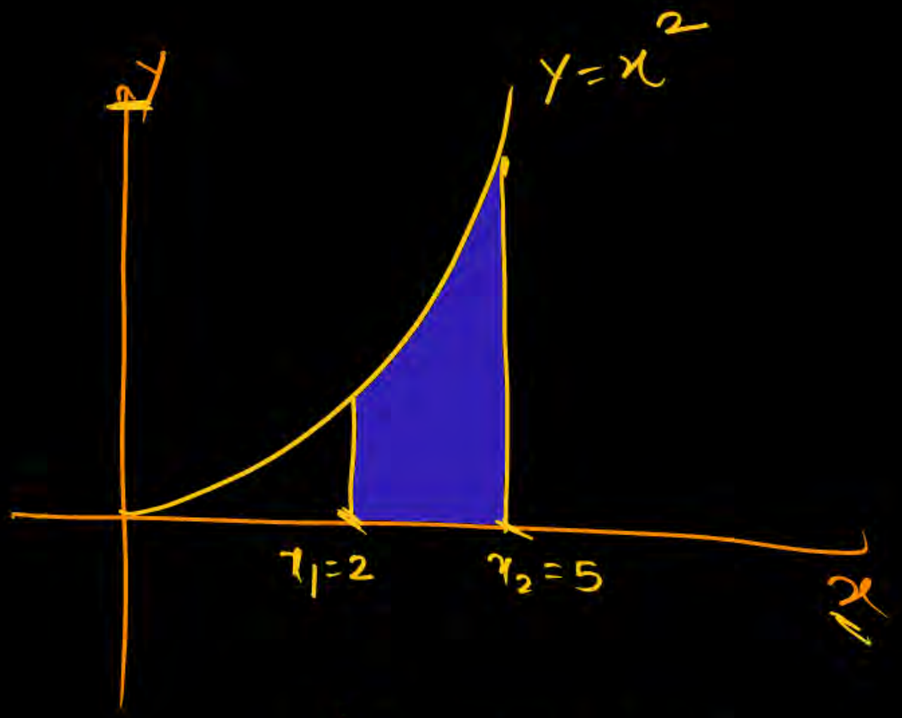
$$\int_{\pi/2}^{2\pi} \sin\theta \, d\theta = -1 \checkmark$$



$\int y\,dx$ = Area of y/x graph

$\int \sin\theta \, d\theta$ = Area of sinθ/θ graph





Area under Curve

$$\int_{r}^{\infty} \frac{G m_1 m_2}{x^2}\, dx = G m_1 m_2 \int_{r}^{\infty} \frac{1}{x^2}\, dx$$

$$= G m_1 m_2 \left[-\frac{1}{x} \right]_{r}^{\infty}$$

$$= -G m_1 m_2 \left[\frac{1}{\infty} - \frac{1}{r} \right]$$

$$= -\frac{G m_1 m_2}{r}$$

Sanyharsh Assignment

do it today

Question

Find the equation of tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

1. $y = \frac{x}{4} + 1$
2. $y = 3x - 2$
3. $y = 5x + 7$
4. $y = 4x + 8$

Question

The differentiation of function $f(x) = (3x + 2)^{3/2}$ w.r.t. x is

1. $\frac{9}{2}(3x + 2)^{1/2}$
2. $\frac{3}{2}(3x + 2)^{1/2}$
3. $\frac{3}{2}(3x + 2)^{-1/2}$
4. $\frac{9}{2}(3x + 2)^{-3/2}$

Question

A boy completes one round of a circular track of radius 20m in 50 seconds. The displacement at the end of 4 minute 10 second will be

1. 40 m
2. 20 m
3. 80 π m
4. Zero

Question

The derivative of function $f(x) = \log_e (2x)$ w.r.t. t is

1. $f'(x) = \frac{1}{2x}\frac{dx}{dt}$
2. $f'(x) = \frac{1}{x}\frac{dx}{dt}$
3. $f'(x) = \frac{1}{x^2}\frac{dx}{dt}$
4. $f'(x) = \frac{1}{2x^2}\frac{dx}{dt}$

Question

If $s = \frac{t}{2t+1}$ then find $\frac{ds}{dt}$

1. $\frac{5}{(4t+2)^2}$
2. $\frac{1}{(2t+1)^2}$
3. $\frac{1}{(3t-2)^2}$
4. $\frac{4}{(2t+1)^3}$

Question

If $x = 2t^3$ and $y = 3t^2$, then value of $\frac{dy}{dx}$ is

1. $\frac{t^2}{2}$
2. $\frac{2}{t^2}$
3. $\frac{1}{t}$
4. $\frac{1}{t^2}$

Question

If $x = 2(\theta + \sin\theta)$ and $y = 2(1 - \cos\theta)$, then value of $\frac{dy}{dx}$ is

1. $\tan\left(\frac{\theta}{2}\right)$
2. $\cot\left(\frac{\theta}{2}\right)$
3. $\sin\left(\frac{\theta}{2}\right)$
4. $\cos\left(\frac{\theta}{2}\right)$

Question

Find the slope of the tangent to the graph $y = x^3 - x^2$ at $x = -1$

1. 4
2. 5
3. 3
4. 6

Question

The displacement x of a particle moving along x-axis at time t is given by $x^2 = 2t^2 + 6t$. The velocity at any time t is:

1. $\frac{4t + 6}{x}$
2. $\frac{2t - x}{x}$
3. $4t + 6$
4. $\frac{2t + 3}{x}$

Question

The position of a particle with respect to time t along y-axis is given by: $y = 12t^2 - 2t^3$, where, y is in metres and t is in seconds. When the particle achieves maximum speed, the position of the particle would be

1. 64 m
2. 40 m
3. 16 m
4. 32 m

Question

$\int_2^4 \frac{1}{x}\,dx$ is equal to

1. $\ln 2$
2. $2 \ln 2$
3. $\ln\left(\frac{3}{2}\right)$
4. $2 \ln 4$

Question

If $y = \log_{10} x$, then the value of $\frac{dy}{dx}$ is

1. $\frac{1}{x}$
2. $\frac{x}{\log_e 10}$
3. $\frac{\log_{10} e}{x}$
4. $\frac{\log_e 10}{x}$

Question

Let $y = x^2 + x$, the minimum value of y is

1. $-1/4$
2. $1/2$
3. $1/4$
4. $-1/2$

Question

If $y = A \sin(\omega t - kx)$, then the value of $\frac{dy}{dx}$ is

1. $A \cos(\omega t - kx)$
2. $-A\omega \cos(\omega t - kx)$
3. $Ak \cos(\omega t - kx)$
4. $-Ak \cos(\omega t - kx)$

Question

$\int_{\infty}^{r} \frac{GM_1m_2}{r^2} dr$ is equal to

1. $\frac{Gm_1m_2}{r}$
2. $-\frac{Gm_1m_2}{r^3}$
3. $\frac{Gm_1m_2}{r^3}$
4. $-\frac{Gm_1m_2}{r}$

Question

If $y = A \sin(\omega t - kx)$, then the value of $\frac{d^2y}{dt^2} / \frac{d^2y}{dx^2}$ is

1. $\frac{k^2}{\omega^2}$
2. $-\frac{k^2}{\omega^2}$
3. $\frac{\omega^2}{k^2}$
4. $-\frac{\omega^2}{k^2}$

Question

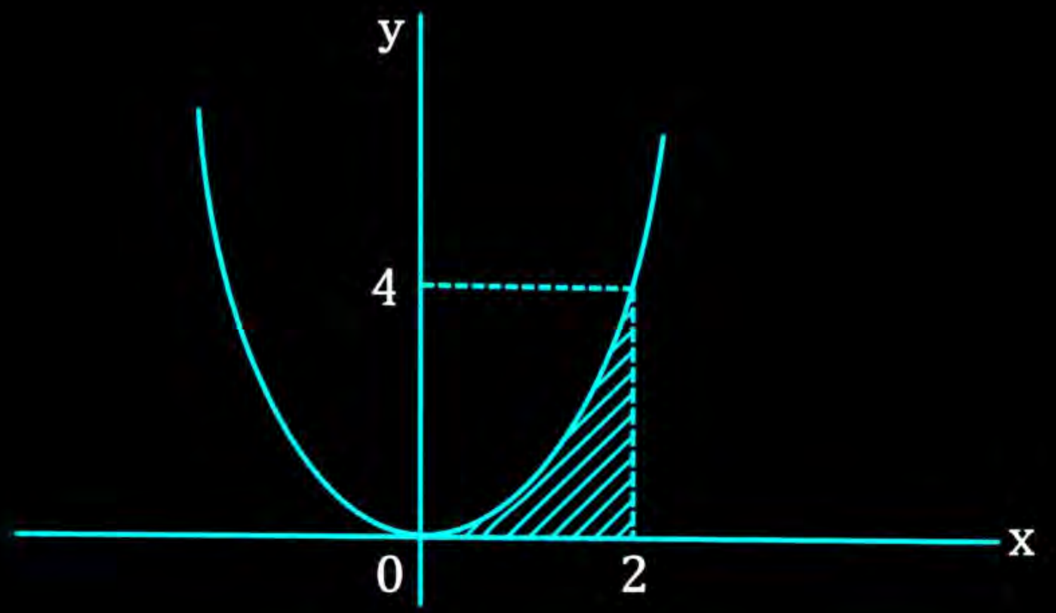
$$\int_0^L \frac{dx}{(ax+b)} =$$

1. $\frac{1}{b}\ln\left(\frac{aL+b}{b}\right)$
2. $-\frac{1}{a}\ln\left(\frac{aL+b}{b}\right)$
3. $\frac{1}{a}\ln\left(\frac{aL+b}{b}\right)$
4. $-\frac{1}{b}\ln\left(\frac{aL+b}{b}\right)$

Question

Consider the parabola $y = x^2$. The shaded area is

1. 2/3
2. 5/3
3. 7/3
4. 8/3



Question

A particle moves in a straight line, so that $x = \sqrt{t}$ then its acceleration is proportional to

1. Velocity
2. $(\text{Velocity})^{3/2}$
3. $(\text{Velocity})^{3}$
4. $(\text{Velocity})^{2}$

Question

Evaluate $\int x^2 . \sin(x^3) . dx$

1. $\frac{\sin x^3}{3} + c$
2. $\cos x^3 + c$
3. $\sin x^3 + c$
4. $\frac{-\cos x^3}{3} + c$

Question

If $F = \frac{2}{\sin\theta + \cos\theta}$ then the minimum value of F out of the following option is

1. 1
2. $\frac{1}{\sqrt{2}}$
3. $\frac{3}{\sqrt{2}}$
4. $\sqrt{2}$

Question

The acceleration of a particle moving along a straight line at any time t is given by $a = 4 - 2v$, where v is the speed of particle at any time t. The maximum velocity is

1. 4 m/s
2. 2 m/s
3. 6 m/s
4. Infinity

Thank You